

AEGIS Project

Stage-N: Hybrid Quantum-Classical Token Execution Layer

Formal Specification for RiftLang Protocol Stack

Version 1.0 - Technical Specification Document

OBINexus Computing Division

Toolchain: `riftlang.exe` → `.so.a` → `rift.exe` → `gosilang`

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1 Executive Summary

This document formalizes Stage-N of the RiftLang Protocol Stack, establishing the critical interface between quantum probabilistic computation and classical deterministic execution. Stage-N enables seamless transitions between quantum superposition states and classical computational models while maintaining strict AEGIS governance compliance throughout the quantum-classical boundary.

The specification defines standardized patterns for stages 0 through N+1 (currently implemented through Stage-7), providing a unified framework for quantum token management, collapse operations, and memory-governed parsing within the RIFT domain-specific language ecosystem.

2 Stage Evolution Framework

Stage Progression Model

- **Stage-0:** Token initialization and classical baseline
- **Stage-1:** Quantum extension introduction
- **Stage-2:** Entanglement protocol establishment
- **Stage-3:** Collapse operator implementation
- **Stage-4:** Memory governance integration
- **Stage-5:** Parser unification
- **Stage-6:** AEGIS phase alignment
- **Stage-7:** Full quantum-classical bridge deployment
- **Stage-N:** Dynamic stage instantiation
- **Stage-N+1:** Future extensibility framework

3 Core Architecture and Purpose

3.1 Fundamental Design Principles

Stage-N serves as the quantum-classical computation bridge within the RIFT DSL execution pipeline. The architecture enables:

1. **Quantum state preservation** during computation
2. **Deterministic resolution** when measurement occurs
3. **Governance enforcement** at state transitions
4. **Memory-bounded execution** with Planck-scale constraints

3.2 Integration Points

```

1  @stage_interface[N] {
2      input_stages: [N-1, quantum_init]
3      output_stages: [N+1, classical_exec]
4      governance_hooks: AEGIS_Phase_I_III
5      memory_model: quantum_foam_temporal
6      compliance_level: STRICT
7  }

```

Listing 1: Stage-N Integration Architecture

4 Quantum Token Specification

4.1 Core Quantum Token Definition

The fundamental quantum token represents a qubit in superposition state with complex amplitude coefficients.

```

1  @token[quantum] qbit superposition( $\alpha$ ,  $\beta$ ) → QINT
2  Where:
3       $\alpha$  in C : Complex amplitude for  $|0\rangle$  state
4       $\beta$  in C : Complex amplitude for  $|1\rangle$  state
5      Constraint:  $|\alpha|^2 + |\beta|^2 = 1$  (normalization)

```

Listing 2: Quantum Token Base Definition

4.2 Extended Quantum Token Attributes

```

1  @token_extension[quantum] {
2      # Dirac notation representation
3      bra_ket_notation:  $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ 
4
5      # Normalization enforcement with tolerance
6      amplitude_norm: enforce( $|\alpha|^2 + |\beta|^2 = 1.0 \pm \epsilon$ )
7      Where:  $\epsilon = 10^{-15}$  (machine  $\epsilon$ )
8
9      # Decoherence threshold at Planck scale
10     decohere_threshold:  $\tau_{\text{planck}} = 5.39 \times 10^{-44}$  seconds
11
12     # Entanglement tracking
13     entanglement_flag: bool
14     entanglement_partners: QINT[] (max_size = 6)
15
16     # Phase coherence bounds
17     phase_coherence:  $\phi$  in  $[0, 2\pi]$ 
18     phase_drift_rate:  $d\phi/dt \leq \pi/\tau_{\text{coherence}}$ 
19 }

```

Listing 3: Quantum Token Extensions

4.3 Quantum Token Memory Alignment

```

1  @memory_align[quantum] {
2      alignment: 8-qubit boundary
3      span_type: distributed_quantum_foam
4      coherence_window: planck_time
5      isolation_level: phase_locked
6
7      layout: {
8          |q0> |q1> |q2> |q3> |q4> |q5> |q6> |q7>
9          [amplitude_real][amplitude_imag]
10         [phase][entangle_mask][coherence]
11     }
12 }
```

Listing 4: Memory Alignment Specification

5 Classical Resolution Operator

5.1 Collapse Operator Definition

The collapse operator manages the quantum-to-classical transition under governance constraints.

```

1  @operator collapse {
2      input: QINT                                # Quantum integer in
3          superposition
4      condition: coherence ≥ PLANCK_THRESHOLD
5      output: INT                                # Classical deterministic
6          integer
7      audit: quantum_event_log                    # Governance audit trail
8
9      properties: {
10         irreversible: true
11         measurement_basis: computational
12         entropy_increase: Δ_S > 0
13     }
14 }
```

Listing 5: Collapse Operator Specification

5.2 Piecewise Collapse Logic

```

1  PROCEDURE quantum_classical_collapse(q: QINT, E: energy, m:
2      mass):
3      LET c2 = speed_of_light2
4      LET E_planck = ħ * ω
5
6      IF E2 ≤ m * c2:
7          # Low energy - classical behavior dominates
```

```

7      state := CLASSICAL_DETERMINISTIC
8      RETURN cast_to_int(q, method="measurement")
9
10     ELIF E > E_critical AND q.coherence ≥  $\tau_{\text{planck}}$ :
11         # High energy with coherence - forced collapse
12         TRIGGER collapse_event {
13             log: quantum_event_log
14             timestamp: current_planck_time
15             method: "forced_decoherence"
16         }
17         state := CLASSICAL_COLLAPSED
18         RETURN probabilistic_cast(q)
19
20     ELSE:
21         # Maintain quantum superposition
22         state := QUANTUM_SUPERPOSITION
23         EVOLVE q WITH hamiltonian(H)
24         RETURN q # Preserve quantum state
25
26     END IF
27 END PROCEDURE

```

Listing 6: Collapse Decision Tree

5.3 State Transition Matrix

```

1  @state_transition_matrix {
2      QUANTUM → CLASSICAL: {
3          trigger: measurement OR decoherence
4          probability:  $|\langle \psi | \phi \rangle|^2$ 
5          governance: collapse_contract
6          audit_level: MANDATORY
7      }
8
9      CLASSICAL → QUANTUM: {
10         trigger: superposition_gate
11         condition: coherence_budget > threshold
12         governance: quantum_init_contract
13         audit_level: STRICT
14     }
15
16     QUANTUM → QUANTUM: {
17         trigger: unitary_evolution
18         operator:  $U = \exp(-iHt/\hbar)$ 
19         governance: evolution_contract
20         audit_level: PERIODIC
21     }
22 }

```

Listing 7: Quantum-Classical State Transitions

6 Memory-Governed Quantum Parser

6.1 Hybrid Token Grammar

The parser must handle both quantum and classical tokens with appropriate type safety.

```

1  @parser[hybrid_quantum_classical] {
2      token_types: {
3          QINT      : quantum_integer[superposition]
4          INT       : classical_integer[deterministic]
5          FLOAT     : classical_float[ieee754]
6          BRA       :  $\langle\psi|$  quantum_state
7          KET       :  $|\psi\rangle$  quantum_state
8          QFLOAT    : quantum_float[superposition]
9      }
10
11     # Regex pattern with quantum extensions
12     parse_rules: R"/([QC])(INT|FLOAT|STATE)/gmi[tb]"
13     Where:
14         g: global matching
15         m: multiline quantum states
16         i: case-insensitive operators
17         t: top-down classical resolution
18         b: bottom-up quantum composition
19 }

```

Listing 8: Token Type Definitions

6.2 Temporal Memory State Management

```

1  @memory_state::quantum_foam {
2      lifetime: planck_time = 5.39 x 10-44 seconds
3      scope: local_superposition
4
5      allocation: {
6          classical_mode: align(4096_bits)
7          quantum_mode: align(8_qubits)
8          hybrid_mode: interleaved_coherent
9      }
10
11     persistence: {
12         coherent_duration:  $\tau_{\text{coherence}}$ 
13         decoherence_rate:  $\Gamma = 1/T_2$ 
14         error_threshold: 10-9
15     }
16
17     governance: {
18         max_entanglement_depth: 6
19         bell_state_limit: 4_pairs
20         gc_policy: phase_aware_collection
21     }

```



```
22 }
```

Listing 9: Quantum Memory Management

6.3 Parser State Machine

```

1  AUTOMATON quantum_parser {
2      states: {S_INIT, S_QUANTUM, S_CLASSICAL, S_COLLAPSE,
3              S_MEASURE}
4
5      transitions: {
6          S_INIT → S_QUANTUM:
7              condition: detect_superposition_token()
8              action: init_quantum_context()
9
10         S_QUANTUM → S_COLLAPSE:
11             condition: coherence < PLANCK_THRESHOLD
12             action: prepare_collapse()
13
14         S_COLLAPSE → S_CLASSICAL:
15             condition: collapse_complete()
16             action: emit_classical_token()
17
18         S_QUANTUM → S_MEASURE:
19             condition: measurement_operator()
20             action: von_neumann_projection()
21
22         S_MEASURE → S_CLASSICAL:
23             condition: measurement_complete()
24             action: emit_measured_value()
25     }
26
27     error_states: {
28         E_COHERENCE_LOST: recovery = forced_collapse
29         E_ENTANGLE_VIOLATION: recovery = isolate_subsystem
30         E_MEMORY_OVERFLOW: recovery = quantum_gc
31     }
32 }
```

Listing 10: Quantum Parser Automaton

7 Governance Constraint Declarations

7.1 Core Governance Rules

```

1  @gov_rule::collapse_contract {
2      requires: {
3          coherence ≥ planck_threshold
4          entanglement_depth ≤ max_allowed
5      }
6  }
```

```

5         quantum_budget > operation_cost
6         audit_trail.enabled = true
7     }
8
9     prohibits: {
10         superposition_state > max_density_matrix_size
11         concurrent_measurements > 1
12         phase_drift >  $\pi/4$ 
13         untracked_entanglement = true
14     }
15
16     audit: {
17         log_destination: quantum_event_log
18         retention_period: 7_stages
19         cryptographic_seal: SHA3-256
20         immutability: blockchain_anchored
21     }
22 }

```

Listing 11: Collapse Contract Governance

7.2 Resource Management Constraints

```

1 @gov_rule::quantum_resource_management {
2     allocation_policy: {
3         max_qubits_per_token: 16
4         max_entangled_pairs: 8
5         decoherence_budget: 1000_planck_times
6         memory_quota: 1MB_quantum_foam
7     }
8
9     cleanup_policy: {
10         auto_collapse_timeout: 100_planck_times
11         garbage_collection: phase_aware
12         memory_reclaim: immediate
13         entanglement_pruning: depth_first
14     }
15
16     cost_model: {
17         superposition_cost: 0.1_per_qubit_per_cycle
18         entanglement_cost: 0.3_per_pair
19         measurement_cost: 0.2_per_operation
20         coherence_maintenance: 0.05_per_planck_time
21     }
22 }

```

Listing 12: Quantum Resource Governance

7.3 Security and Validation Rules

```

1  @gov_rule::quantum_security {
2      validation: {
3          state_vector_normalization: continuous
4          no_cloning_enforcement: strict
5          basis_state_verification: periodic(10_cycles)
6          bell_inequality_check: on_entanglement
7      }
8
9      access_control: {
10         quantum_state_read: privileged_only
11         collapse_trigger: authorized_operators
12         entanglement_create: rate_limited(10/sec)
13         phase_manipulation: governance_approved
14     }
15
16     integrity: {
17         checksum_algorithm: quantum_hash_SHA3Q
18         tamper_detection: bell_inequality_test
19         audit_trail: immutable_quantum_ledger
20         replay_protection: nonce_per_operation
21     }
22 }

```

Listing 13: Quantum Security Governance

8 Integration with AEGIS Phase Architecture

8.1 Phase I - Matrix Parity Integration

```

1  INTEGRATION matrix_parity_bridge {
2      quantum_to_fft: {
3          INPUT: QINT[superposition]
4          PROCESS:
5              1. Extract amplitude vectors ( $\alpha$ ,  $\beta$ )
6              2. Map to FFT basis:  $F(|\psi\rangle) = \text{Sum}(\alpha_k * e^{(2*\pi*ijk/N)})$ 
7              3. Apply parity constraints from Phase I
8              4. Verify matrix eigenvalue stability
9          OUTPUT: FFT_MATRIX[classical]
10     }
11
12     governance: {
13         parity_check: R"/[01]{8}/g"
14         matrix_alignment: 8x8_quantum_block
15         eigenvalue_threshold:  $|\lambda| < 1.0$ 
16     }
17 }

```

Listing 14: Matrix Parity Bridge

8.2 Phase II - Token Stream Management

```

1  INTEGRATION token_stream_quantum {
2      stream_mode: {
3          classical: sequential_ordered
4          quantum: parallel_superposed
5          hybrid: context_switched
6      }
7
8      synchronization: {
9          barrier: quantum_measurement_point
10         ordering: causal_cone_preservation
11         latency: ≤ coherence_window
12     }
13
14     buffering: {
15         quantum_buffer: circular_phase_locked
16         classical_buffer: FIFO_deterministic
17         transition_buffer: copy_on_collapse
18     }
19 }

```

Listing 15: Token Stream Integration

8.3 Phase III - Planck Verification

```

1  INTEGRATION planck_verification {
2      collapse_window: {
3          detection: coherence < PLANCK_THRESHOLD
4          action: enforce(collapse_contract)
5          verification: cryptographic_proof
6          timing: exact_planck_time
7      }
8
9      quantum_classical_boundary: {
10         transition_log: {
11             timestamp: planck_time_resolution
12             state_before:  $|\psi\rangle$ 
13             state_after: classical_value
14             entropy_change:  $\Delta_S$ 
15             information_preserved:  $I = -\text{Sum}(p \log p)$ 
16         }
17     }
18
19     entanglement_boundary: {
20         max_distance: 6_hops
21         isolation_enforcement: bell_state_collapse
22         audit_requirement: full_trace
23         correlation_preservation: EPR_compliant
24     }
25 }

```

Listing 16: Planck Scale Verification

9 Runtime Execution Model

9.1 Dual-Mode Execution Pipeline

```

1 PIPELINE rift_stage_n_execution {
2     MODE classical {
3         stages: tokenize → parse → validate → execute
4         memory: sequential_4096_aligned
5         concurrency: mutex_protected
6         error_handling: exception_based
7     }
8
9     MODE quantum {
10        stages: superpose → entangle → evolve → measure
11        memory: distributed_8qubit_foam
12        concurrency: phase_locked_parallel
13        error_handling: decoherence_recovery
14    }
15
16    MODE hybrid {
17        stages: detect_context → switch_mode → process →
18            reconcile
19        memory: adaptive_alignment
20        concurrency: quantum_classical_barrier
21        error_handling: graceful_degradation
22    }
23
24    performance: {
25        classical_throughput: 106 ops/sec
26        quantum_coherence: 1000 x planck_time
27        transition_overhead: < 1us
28    }
29 }
```

Listing 17: Stage-N Execution Pipeline

9.2 Context Switching Protocol

```

1 PROTOCOL context_switch {
2     classical_to_quantum: {
3         save_classical_state()
4         init_quantum_registers()
5         prepare_superposition()
6         verify_coherence()
7         enable_quantum_operations()
8     }
9 }
```

```

8     }
9
10    quantum_to_classical: {
11        prepare_measurement()
12        collapse_superposition()
13        extract_classical_value()
14        cleanup_quantum_resources()
15        restore_classical_context()
16    }
17
18    switch_overhead: {
19        time_cost: 0(n_qubits)
20        memory_cost: 0(2^n_qubits)
21        governance_cost: 0(audit_depth)
22    }
23 }

```

Listing 18: Quantum-Classical Context Switch

10 Example Usage Patterns

10.1 Basic Quantum Token Operations

```

1  # Initialize quantum token in superposition
2  @quantum
3  let q_value: QINT = superposition(0.707, 0.707) # |+> state
4
5  # Entangle two quantum tokens
6  @quantum
7  let q_pair: (QINT, QINT) = entangle(q_value, q_other)
8
9  # Conditional collapse based on coherence
10 @hybrid
11 if coherence(q_value) < PLANCK_THRESHOLD {
12     let classical_result: INT = collapse(q_value)
13     process_classical(classical_result)
14 } else {
15     evolve_quantum(q_value, hamiltonian)
16 }
17
18 # Measurement with basis selection
19 @quantum
20 let measured: INT = measure(q_value, basis="X")

```

Listing 19: Quantum Token Usage Examples

10.2 Hybrid Computation Pattern

```

1  @hybrid_algorithm grover_search {

```

```

2      # Classical preprocessing
3      let dataset: INT[] = load_classical_data()
4      let target: INT = define_search_target()
5
6      # Quantum acceleration phase
7      @quantum {
8          let q_register: QINT[] = init_superposition(size=log2(
9              dataset.length))
10
11          repeat sqrt(dataset.length) times {
12              apply_oracle(q_register, target)
13              apply_diffusion(q_register)
14
15              if coherence_degraded(q_register) {
16                  refresh_quantum_state(q_register)
17              }
18
19          let result_index: INT = measure_all(q_register)
20      }
21
22      # Classical verification
23      @classical {
24          verify_result(dataset[result_index], target)
25          return dataset[result_index]
26      }
27 }

```

Listing 20: Hybrid Quantum-Classical Algorithm

11 Formal Verification Properties

11.1 Safety Properties

```

1  PROPERTY quantum_safety {
2      # G1: Normalization is always maintained
3      [] (forall q: QINT . |amplitude(q)|2 = 1.0 +/- ε)
4
5      # G2: No-cloning theorem is preserved
6      [] (not exists operation . clone( |ψ⟩ ) → |ψ⟩ |ψ⟩ )
7
8      # G3: Causality is respected
9      [] (forall measurement . timestamp(cause) < timestamp(
10          effect))
11
12     # G4: Measurement irreversibility
13     [] (collapsed(q) → not quantum(q))
14 }

```

Listing 21: Quantum Safety Invariants

11.2 Liveness Properties

```

1  PROPERTY quantum_liveness {
2      # L1: Every quantum state eventually decoheres
3      <> (forall q: QINT . coherence(q) < PLANCK_THRESHOLD)
4
5      # L2: Measurements eventually complete
6      (measure_initiated(q) → <> measure_complete(q))
7
8      # L3: Resources are eventually reclaimed
9      []<> (allocated_qubits = deallocated_qubits)
10
11     # L4: Entanglements eventually resolve
12     <> (forall entangled_pair . separated OR measured)
13 }

```

Listing 22: Quantum Liveness Guarantees

12 Performance Specifications

Performance Requirements

- **Quantum Coherence Time:** $\geq 1000\tau_{planck}$
- **State Preparation:** < 10 ns per qubit
- **Measurement Latency:** < 100 ns
- **Context Switch Overhead:** $< 1\mu s$
- **Memory Efficiency:** $O(n)$ for n qubits
- **Entanglement Depth:** Maximum 6 levels
- **Error Rate:** $< 10^{-9}$ per operation

13 Firmware Integration Guidelines

13.1 Git-RAF Integration Points

```

1  @firmware_integration {
2      git_raf_hooks: {
3          pre_commit: validate_quantum_governance()
4          post_merge: verify_collapse_consistency()
5          pre_push: audit_quantum_operations()
6      }
7
8      versioning: {
9          quantum_state_snapshot: on_commit
10         collapse_history: immutable_log

```



```

11         entanglement_graph: version_tracked
12     }
13
14     deployment: {
15         stage: [0..N+1]
16         firmware_target: quantum_coprocessor
17         validation_level: AEGIS_COMPLIANT
18     }
19 }

```

Listing 23: Git-RAF Firmware Hooks

14 Conclusion and Future Extensions

Stage-N of the RiftLang Protocol Stack provides a robust foundation for hybrid quantum-classical computation within the AEGIS governance framework. The specification enables:

- Seamless quantum-classical transitions
- Governance-compliant state management
- Planck-scale temporal constraints
- Integration with existing AEGIS phases
- Extensibility for future quantum algorithms

Future stages (N+1 and beyond) will extend this framework to support:

- Distributed quantum computation
- Fault-tolerant quantum error correction
- Advanced entanglement protocols
- Quantum machine learning integration

A Glossary of Terms

QINT Quantum Integer - A quantum register holding superposition states

Planck Time $\tau_{planck} = 5.39 \times 10^{-44}$ seconds

Coherence Quantum state phase relationship preservation

Collapse Quantum to classical state transition

AEGIS Automated Enterprise Governance Intelligence System

B References

1. AEGIS Project Technical Specification v2.0
2. RiftLang Compiler Documentation
3. Quantum Computing Governance Framework
4. Git-RAF Firmware Integration Manual
5. OBINexus Toolchain Architecture Guide